

TADIKAMALLA HARSHVARDHAN GUPTA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate focused on AI automation, backend engineering, and LLM-powered systems. Built local/fine-tuned LLM document workflows, RAG chatbots, multi-agent lead discovery platforms, and backend CRM automation using Python, FastAPI/Flask, SQL/MongoDB, LangChain/LangGraph, and vector databases. Internship experience includes automation engineering at Rolls-Royce Power Systems and backend development for investor CRM workflows.

EDUCATION

SRM Institute of Science and Technology, Chennai

B.Tech in Computer Science (Core)

2022 - 2026 (Expected)

- CGPA: 8.04/10(till 7th semester)
- Relevant Coursework: Database Management, Machine Learning, Software Engineering, Data Structures, Algorithms

TECHNICAL SKILLS

Languages:	Python, SQL, JavaScript/TypeScript
Backend:	Flask, FastAPI, REST APIs, Streamlit, Socket.IO, Uvicorn
AI/LLM:	Local LLMs, Fine-Tuning, RAG, LangChain, LangGraph, Ollama, Vector Databases
Data & ML:	Web Scraping, ETL, Pandas, NumPy, Scikit-learn, TensorFlow, Time-Series Analysis
Databases:	PostgreSQL, MySQL, SQLite, MongoDB, ChromaDB
Tools & Platforms:	Git, GitHub, GitHub API, Docker, Postman, Excel, CI/CD
Concepts:	Multi-Agent Systems, API Integration, Data Pipelines, Automation, System Design

PROFESSIONAL EXPERIENCE

Rolls-Royce Power Systems

Automation Engineer Intern

July 2025 - December 2025

- Built DocAI, a local document intelligence tool for uploading PDFs, Excel sheets, and Word documents, then querying/extracting information using a fine-tuned local LLM workflow.
- Implemented Python extraction pipelines with PDF/text parsing, table extraction, OCR fallback, structured JSON outputs, and batch document processing.
- Developed Python test automation for DevOps CI/CD validation and automated reporting workflows using Python, SQL, and Excel.

Contisoft Technologies

Software Engineer Intern

November 2024 - February 2025

- Built Stamp Paper OCR automation for Indian stamp papers, extracting certificate/reference numbers, denomination, state, and validation signals from images and PDFs.
- Developed an auto-learning web scraping RAG chatbot with website crawling, content chunking, vector storage, and local LLM-based question answering.
- Implemented Flask/FastAPI services, API integrations, validation, and automated Excel/report generation.

HustleBee Software Private Limited

Full Stack Developer - Intern

January 2026 - April 2026

- Developed backend modules for an investor CRM platform, including lead/contact workflows, REST APIs, database operations, and automation logic.
- Built Lead Force AI, a multi-agent lead discovery and enrichment system that searches public web sources, extracts lead signals, and organizes prospects for CRM usage.
- Implemented lead ingestion, filtering, deduplication, API integrations, logging, and deployment-ready backend improvements.

PROJECTS

LeadForge AI - Multi-Agent B2B Lead Discovery Platform

Tech Stack: FastAPI, LangGraph, LangChain, React, Vite, Pydantic, API Integrations

- Built a multi-agent lead discovery platform that converts a product description into a prioritized B2B prospecting report.
- Orchestrated research, deduplication, verification, demand scoring, enrichment, and synthesis workflows across public and enrichment data sources.

GitHub: <https://github.com/harsh20048/LeadForge-AI>

Auto-Learning Web Scraping RAG Chatbot

Tech Stack: Python, Flask, BeautifulSoup, ChromaDB, sentence-transformers, Ollama/Mistral

- Built a self-learning website chatbot that crawls pages, chunks content, stores embeddings, and answers questions using local LLM-based RAG.
- Exposed APIs for scraping, scrape-status tracking, chatbot queries, chat-history export, and local model error handling.

GitHub: <https://github.com/harsh20048/web-scraping-chatbot>

Codebase Onboarding Agent - AI Repository Understanding Tool

Tech Stack: Python, LangChain, GitHub API, Vector Databases, Streamlit

- Built an AI agent that analyses GitHub repositories and generates structured onboarding notes for architecture, workflows, and key modules.
- Used repository ingestion, semantic search, and LLM summarization to help developers understand unfamiliar codebases faster.

GitHub: <https://github.com/harsh20048/codebase-onboarding-agent>

PATENT & RESEARCH

Semantic Compression of Chain-of-Thought Reasoning in Large Language Models

Patent Application Filed

- Proposed a semantic compression framework to reduce LLM reasoning/inference cost while preserving output quality for enterprise LLM workflows.
- Explored compressed intermediate reasoning traces for cost-sensitive document automation and AI systems.

ACHIEVEMENTS & CERTIFICATIONS

- Top 5/99 teams - ACE HACK Hackathon (built an AI-powered solution for a real-world problem)
- Gold Medalist - National Karate Championship
- Black Belt - Taekwondo